

NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

Patient ID: IND-KA-PHC-8922 **Age/Sex:** 61 / Male **Screening Date:** 05-Sep-2026 08:42
PHC Health Centre: PHC Mulbagal, Kolar District **Frontline Operator:** ASHA Worker #108

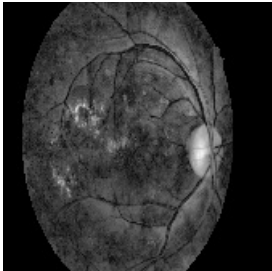
TRIAGE ACTION: URGENT SPECIALIST REFERRAL REQUIRED (<48 HOURS)

High risk of irreversible vision loss. Tertiary Vitreoretinal evaluation mandatory.

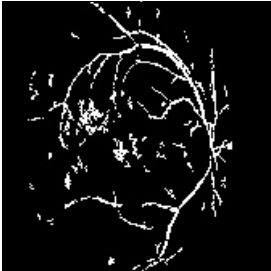
1. Raw Fundus Scan



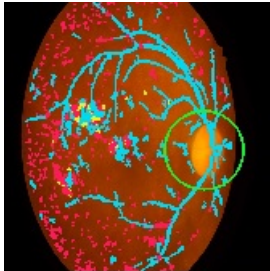
2. Enhanced Green



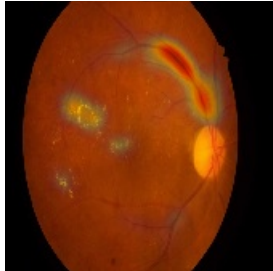
3. Retinal Vessels



4. Lesion Detections



5. Grad-CAM XAI

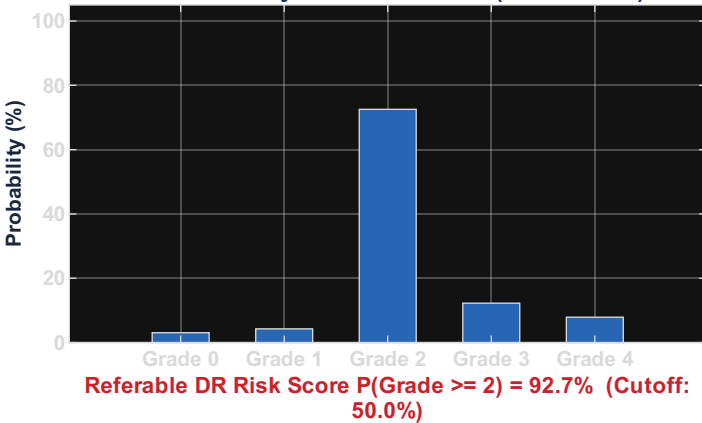


Legend: [Cyan]: Vascular Tree | [Yellow]: Hard Exudates (Lipids) | [Magenta]: Microaneurysms/Hemorrhages | [Jet Colormap]: AI Focus Regions

QUANTITATIVE BIOMARKER ANALYSIS

Image Quality Assessment (IQA): PASS (BRISQUE: 35.3)
Mean Illumination Level: 30.0% (Dynamic Range: 0.19)
Vascular Density (Coverage): 9.52% of FOV
Hard Exudates (Bright Lesions): 28 clusters (608 px area)
Microaneurysms (Early NPDR): 447 detected foci
Retinal Blot Hemorrhages: 15 detected sites
Optic Disc Status: Localized (Center: [380, 282])

AI DR Severity: Moderate NPDR (Conf: 72.5%)



{bf OPTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Time required: under 30 seconds)}

Clinical Impression: ☐ Concordant with AI diagnosis ☐ Upgrade Severity ☐ Downgrade / Artefact

Action Prescribed: ☐ Tele-Consultation Booked ☐ Anti-VEGF / Laser Referral ☐ PHC Annual Rescreen

Reviewing Ophthalmologist Signature: _____ Date/Time: _____